The Chinese University of Hong Kong, Shenzhen
School of Data Science

MAT3007 · Homework 2

Due: 11:59, June 23

Instructions:

- Homework problems must be carefully and clearly answered to receive full credit. Complete sentences that establish a clear logical progression are highly recommended.
- You must submit your assignment in Blackboard. If some problems need coding, you should paste your code in the file. Please upload only one file(pdf). The file name should be in the format **last name-first name-hw2. Any nonstandard assignment will not be graded.**.
- The homework must be written in English.
- Late submission will not be graded.
- Each student **must not copy** homework solutions from another student or from any other source.

Problem 1 (20pts). Reformulate NLP as LP

Reformulate the following problems as linear programming

$$\begin{aligned} \text{minimize} \quad & x_2 + |2x_1 - x_3| \\ \text{s.t.} \quad & |x_1 + 3| + |x_2| \leq 6 \\ & x_3^2 \leq 4 \end{aligned}$$

Please also write down its standard form.

Solution.

Let $y_1 = |2x_1 - x_3|, y_2 = |x_1 + 3|, y_3 = |x_2|$.

$$\begin{aligned} \text{minimize} \quad & x_2 + y_1 \\ \text{s.t.} \quad & y_1 \geq 2x_1 - x_3 \\ & y_1 \geq -2x_1 + x_3 \\ & y_2 + y_3 \leq 6 \\ & y_2 \geq x_1 + 3 \\ & y_2 \geq -x_1 - 3 \\ & y_3 \geq x_2 \\ & y_3 \geq -x_2 \\ & x_3 \leq 2 \\ & x_3 \geq -2 \end{aligned}$$

The standard form:

$$
\begin{array}{ll}
\text{minimize} & x_2 + y_1 \\
\text{s.t.} & y_1 - 2\left(x_1^+ - x_1^-\right) + \left(x_3^+ - x_3^-\right) - s_1 = 0 \\
& y_1 + 2\left(x_1^+ - x_1^-\right) - \left(x_3^+ - x_3^-\right) - s_2 = 0 \\
& y_2 + y_3 - 6 + s_3 = 0 \\
& y_2 - \left(x_1^+ - x_1^-\right) - 3 - s_4 = 0 \\
& y_2 + \left(x_1^+ - x_1^-\right) + 3 - s_5 = 0 \\
& y_3 - \left(x_2^+ - x_2^-\right) - s_6 = 0 \\
& y_3 + \left(x_2^+ - x_2^-\right) - s_7 = 0 \\
& \left(x_3^+ - x_3^-\right) - 2 + s_8 = 0 \\
& \left(x_3^+ - x_3^-\right) + 2 - s_9 = 0 \\
& y_1, y_2, y_3, x_1^+, x_1^-, x_2^+, x_2^-, x_3^+, x_3^- \geq 0 \\
& s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9 \geq 0
\end{array}
$$

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Problem 2 (35pts). Basic solutions and basic feasible solutions
Consider the following linear optimization problem:

$$
\begin{array}{ll}
\text{maximize} & x_1 + 2x_2 + 5x_3 \\
\text{s.t.} & x_1 + x_2 \leq 15 \\
& x_2 + 3x_3 \leq 12 \\
& x_1, x_2, x_3 \geq 0
\end{array}
$$

(a) Transform it into standard form;

(b) Argue without solving this LP that there must exist an optimal solution with no more than 2 positive variables;

(c) List all the basic solutions and basic feasible solutions (of the standard form);

(d) Find the optimal solution by using the results in step (c).

Solution.

(a) Standard form:

$$
\begin{array}{ll}
\text{minimize} & -x_1 - 2x_2 - 5x_3 \\
\text{s.t.} & x_1 + x_2 + x_4 = 15 \\
& x_2 + 3x_3 + x_5 = 12 \\
& x_1, x_2, x_3, x_4, x_5 \geq 0
\end{array}
$$

(b) First, we will argue that there exist an optimal solution for this problem. From the constraints we know the feasible set is bounded. The objective function is to maximize the sum of bounded variables, so an optimal solution must exist. Moreover, since all the rows of the constraint matrix are linearly independent with m = 2 and n = 5, there must exists an optimal solution with no more than 2 positive variables (basic feasible optimal solutions).

(c) $B = \{i \mid x_i > 0\}$.See the following table for the basic solutions and basic feasible solutions:

B	$\{1,2\}$	$\{1,3\}$	$\{1,5\}$	$\{2,3\}$	$\{2,4\}$	$\{2,5\}$	$\{3,4\}$	$\{4,5\}$
x_B	$(3,12)$	$(15,4)$	$(15,12)$	$(15,-1)$	$(12,3)$	$(15,-3)$	$(4,15)$	$(15,12)$
Obj. Val.	-27	-35	-15	-25	-24	-30	-20	0
BFS	Y	Y	Y	N	Y	N	Y	Y

(d) The optimal solution is $(15, 0, 4, 0, 0)$.

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Problem 3 (45pts). A Robust LP Formulation
In this exercise, we consider the following optimization problem:

$$\min_{x \in \mathbf{R}^n} \; c^\top x \quad \text{subject to} \quad \|Ax - b\|_\infty \leq \delta, \quad x \geq 0 \tag{1}$$

where $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$, $c \in \mathbf{R}^n$, and $\delta \geq 0$ are given and $\|y\|_\infty = \max_{1 \leq i \leq p} |y_i|$ denotes the infinity norm of a vector $y \in \mathbf{R}^p$. In the case $\delta = 0$, problem (1) coincides with the standard form for linear programs. The choice $\delta > 0$ can be useful to model situations where A and/or b are not fully or exactly known, e.g., when A and/or b contains certain uncertainty (can be caused by noise). In this case, problem (1) belongs to the so-called robust optimization.

(a) We now consider a specific application of problem (1).

The fruit store in Pandora is producing two different fruit salads A and B. The smaller fruit salad A consists of "1/4 mango, 1/8 pineapple, 3 strawberries"; the larger fruit salad B consists of "1/2 mango, 1/4 pineapple, 1 strawberry". The profits per fruit salad and the total number of fruits in stock are summarized in the following table:

	Mango	Pineapple	Strawberry	Net profit
Fruit salad A	1/4	1/8	3	10 RMB
Fruit salad B	1/2	1/4	1	20 RMB
Stock / Resources	25	10	120	

Suppose all fruits need to be processed and *completely used* to make the fruit salads A and B. Given these constraints, formulate a linear program to maximize the total profits of the fruit store. Show that this program can be expressed in a linear programming problem. In addition, is this linear programming feasible?

Note: Since we want to produce "complete fruit salads", the variables x_1 and x_2 should actually be modeled as integer variables: $x_1, x_2 \in \mathbb{Z}$. However, since we do not know how to deal with these integer constraints in general at this moment, you may just ignore the integrity of x_1, x_2.

(b) We now consider a variant of the problem in part (a). In practice, we allow the use of fruits to have a certain amount of surplus or excessive. All fruits must satisfy the constraint of having at most 5 units left or 5 additional units used. For example, the constraint for Mango can be formulated as:

$$\left|\frac{x_1}{4} + \frac{x_2}{2} - 25\right| \leq 5$$

- Given these constraints, formulate the problem as the form of the robust formulation (1). Then rewrite it into a linear programming problem.
- Sketch the feasible set of this problem.
- Solve the problem graphically, i.e., Calculate the optimal value and the optimal solution set.
- Which constraints are active in the solution?
- Find one integer solution of this problem.

Solution.

(a) Let x_1 and x_2 denote the number fruit salads A and B, respectively. Since all fruits need to be processed completely, we need to satisfy the equality constraints

$$\frac{1}{4}x_1 + \frac{1}{2}x_2 = 25, \quad \frac{1}{8}x_1 + \frac{1}{4}x_2 = 10, \quad 3x_1 + x_2 = 120$$

and the total profit is given by $10x_1 + 20x_2$. This yields the following linear program

$$\min_x \ -10x_1 - 20x_2 \quad \text{s.t.} \quad \begin{cases} x \geq 0, \\ x_1 + 2x_2 = 100, \\ x_1 + 2x_2 = 80, \\ 3x_1 + x_2 = 120. \end{cases}$$

This linear programming is not feasible, as the first two constraints are not consistent.

(b) The robust formulation is given by:

$$\max_x \ 10x_1 + 20x_2 \quad \text{s.t.} \quad \begin{cases} x \geq 0, \\ \left|\frac{x_1}{4} + \frac{x_2}{2} - 25\right| \leq 5, \\ \left|\frac{x_1}{8} + \frac{x_2}{4} - 10\right| \leq 5, \\ |3x_1 + x_2 - 120| \leq 5. \end{cases}$$

Expanding the abs value, we obtain

$$\max_x \ 10x_1 + 20x_2 \quad \text{s.t.} \quad \begin{cases} x \geq 0, \\ x_1 + 2x_2 - 100 \leq 20, \\ x_1 + 2x_2 - 100 \geq -20, \\ x_1 + 2x_2 - 80 \leq 40, \\ x_1 + 2x_2 - 80 \geq -40, \\ 3x_1 + x_2 - 120 \leq 5, \\ 3x_1 + x_2 - 120 \geq -5 \end{cases}$$

We can first draw the feasible region of this problem, which is shown in the Figure 1.

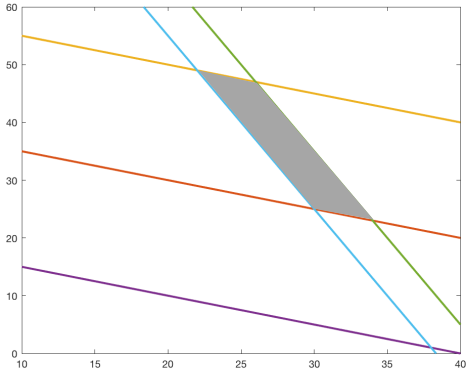


Figure 1: Feasible region

In order to get the optimal solution set and optimal value, we depict the contour lines $\{x : 10x_1 + 20x_2 = c\}$ for the several values of c. i can be seen whe $c = 1200$, the contour line will touch the feasible region from above and achieves the maximum value, 1200; depicted as dashed line in the Figure 2:

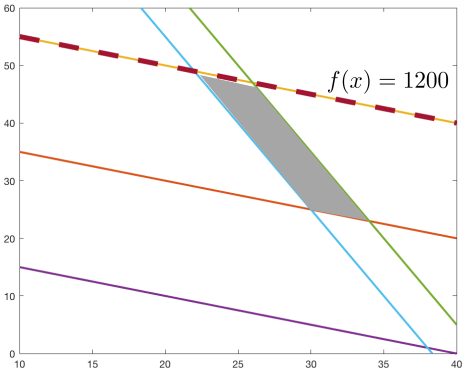


Figure 2: The solution set and optimal value

Furthermore, the optimal solution is not unique and they form a set:

$$X^* := \{x : x_1 + 2x_2 = 120,\ x_1 \in [22, 26]\}.$$

Thus, at an optimal solution $x \in X^*$ either the constraints $x_1 + 2x_2 = 120$ or $x_1 + 2x_2 = 120$ and $3x_1 + x_2 = 115$ or $x_1 + 2x_2 = 120$ and $3x_1 + x_2 = 125$ are active.

As for integer solution, we can let $x_1 \in \{22, 23, 24, 25, 26\}$, we will get $x_2 \in \{49, 48.5, 48, 47.5, 47\}$, respectively. Thus, we can select any of the following plans to achieve the maximum value 1200:

$$(x_1, x_2) = (22, 49), \text{ or, } (x_1, x_2) = (24, 48), \text{ or, } (x_1, x_2) = (26, 47).$$

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